

ML Based Fruit Health and Quality Analysis System

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Abstract—With the constant rise in demand for better quality produce around the world, qualitative assessment of fruits based on the subjective decision-making of assessing inspectors on physical inspection has proved insufficient. The concern of the consumer is always fruit quality. This is the initiative that would build the economic growth of the country, called Automation. In addition to that, it helps in small, micro, medium and large industry as well. The proposed system utilizes image processing to classify and grade the quality of fruits based on features like color, shape etc. so as to differentiate amongst the given fruit. This ranks it as one of the major exporting countries for fruits. Diseases of dirty Fruits and vegetables, Since the growing incidence does not distance fruit spreads from one enlightened state to form a type, most fruits spread in developing countries had very high incidence. **Sensory Characters: Fruit appearance (color, shape, size etc.)**

Keywords—KNN, Machine learning, Quality evaluation, Fruit Quality.

I. INTRODUCTION

The assessment of fruit freshness and quality would become crucial in guaranteeing food safety, enhanced consumer satisfaction, and increased agricultural productivity. Due to this fact, new methods have been developed, and among the new approaches is that which utilizes machine learning techniques. learning, on the other hand is an application of one of the subsets of artificial intelligence that provides a powerful capability in the analysis and processing of big data. Thus, using learned algorithms from data, we can create systems that yield accurate, consistent, and objective measures of fruit quality and health, including color, texture, firmness, or other sensory characteristics, as well as diseases or imperfections not immediately detectable by the naked eye. The sorting machines are comprised of a mechanical sorter that has conveyer belts attached to them. The machine has graded/frayed them on weight, size, shape. If we place a product any fruit, on the rotating conveyer around a machine that has inputs of size, shape etc., and a sorter begin to activate the process of sorting fruits which there is no reference in regard to this sense discussed threshold as each might have their own criteria. To be exact, these threshold values are machine scanned for each single fruit product or sometimes a batch of fruits to ensure quality. Therefore, this method is faster and more accurate in sorting [1,2].

II. LITERATURE STUDY

This review article [3] focuses on the important role that machine learning algorithms and fundamentals could play in protecting a crucial aspect such as food safety and preservation of perishable produce. Therefore it is focusing on the post-harvest safety in fruits and vegetables, as these are considered to be more delicate objects for spoilage. In addition to their specific applications like spoilage/contamination detection (e.g., filtrate with inaccurate salt levels) or storage optimization, it also provides some ML algorithms including SVM, ANN, KNN and Random Forests. Some of these instruments are used to predict shelf life and classify freshness with respect to perishable items, as well as monitor environmental factors such as temperature and humidity. The authors quote even the challenges of ML are the considered nice introduced quality.

[4] leans toward the side of shelf life estimation, quality assessment being largely presented in this case with an experimentation focused on dates they were to develop a non-destructive shelf life indicator through the use machine learning and multispectral sensing. These include studies of different packaging techniques (ambient freshly, vacuum sealed bag, and modified atmosphere packaging) along with the quality parameters (pH, moisture percentage, sugar percentage and tannin concentration). Quality estimation is performed in real-time using Tiny Machine Learning models created and deployed on a low-cost Arduino Nano device based on Spectral data collected by an AS7265x multispectral sensor.

This paper [5] ascertains if machine learning can be applied for the automatic assessment of fruit quality from images. The researchers used some machine learning algorithms that classify the fruit images on qualifying conditions such as ripeness, defects, and other conditions for dealing with the problems of not being invasive and the need for real-time control of fruit quality. It increases efficiency in quality checks without involving all those very complex manual inspections but ensures accuracy and consistency in quality checks. These systems expose problems in training models caused by variations in lighting and fruit appearances but can easily and effectively make quality control more scalable so that food waste is reduced and supply chains optimized.

[6] discuss the possibility of using ML techniques for detecting and sorting various types of fruits and vegetables. Enough literature can be seen that discusses how automation in food processing is significant as it not only saves labor but is effective too, costs are reduced in terms of labor, and results concerning sorting are exact as compared to other human-based options. Applications of a range of ML algorithms have been made to images in order to classify images by shape, color, or texture through applying image processing tools such as k-NN, SVM, and neural networks. Systems whose quality varies and are prohibitively expensive point out a challenge the algorithms also have to be faster and more robust. The scope of the application of this working area is likely to focus on deep learning applications in the future. Improved accuracy and flexibility in real life sorting machines are yet to be devised.

[7] reviewed the potential application of machine learning and unmanned aerial system multispectral imagery in predicting the antioxidant activity of cherry fruits. Literature has expressed the role of antioxidants in health and discussed how remote sensing technology, such as UAS, is changing the essence of agricultural monitoring through the provision of timely data on crop quality. The traditional biochemical techniques for the determination of antioxidants are destructive and time-consuming. This study focuses on the possibility that ML models trained on spectral data can provide a non-destructive efficient alternative in the determination of antioxidant levels in fruits.

[8] have shown the various types of ML techniques deployed in agriculture regarding banana crops. The paper enables appropriate in-depth discussion regarding the scope of improvement in agricultural processes in terms of disease identification, yield prediction, control of pests, and optimization of resources by ML algorithms. This literature review will demonstrate that, indeed, supervised learning methods applied within classifications and predictions are largely represented with decision trees and support vector machines, while unsupervised learning techniques seemed to be applied upon crops data, with the purposes of recognizing patterns. Deep models toward image recognition in identification of pests and diseases is widely applied lately. It identifies the challenges in applying ML in agriculture, such as the substantial datasets required; variability in the environment, and inconsistency in the data labeling. In general, this study supports the possibility of bringing revolution in banana farming by improvements in efficiency, better management of diseases, and smarter agricultural practices by suggesting future directions for overcoming the present challenges in research.

The research study [9] designed a machine vision system for the quality evaluation of dried white mulberry (*Morus alba* L.) moves towards raising awareness about the increasing health benefits provided by this fruit. Especially because the quality of dried mulberries varies with samples and is not consistent, it requires knowledge-based techniques in image processing and artificial intelligence to design a grading tool that can work right. Using a digital camera, they captured images of good and bad mulberries and extracted color and texture features from the images. The best combination of color and texture features that had good classification results for mulberry images was determined using ANN and SVM classifiers. Quadratic sequential feature selection identified 23 as the optimal features. That

is, the system could display an impressive correct classification rate of 100%. Therefore, it presents a promising tool for quality monitoring in the food industry with characteristics of being cost-effective, rapid and intelligent.

The review article [10] discusses the application of machine learning methods to enhance nondestructive methods for the assessment of food quality and safety. The authors go on to discuss different applications of ML, enhanced computer vision, spectroscopy, and sensor technologies applied to improve the nondestructive sensing of properties like freshness in foods. Key contributions pertain to developing image analysis for surface defect identification, hyperspectral imaging for chemical composition, and the development of smart sensors for real-time quality monitoring. The paper outlines various benefits identified: benefits accruing from the application of ML to improvement in terms of accuracy and efficiency. However, some challenges include large datasets and model generalization. The bottom line remains that it means ML is soon to transform and revolutionize food quality assessment in industries.

In paper [11], a review is conducted with regard to an investigation of the application of ML for enhancing non-destructive means for gauging food quality and safety. The paper reviews many applications of ML, computer vision, spectroscopy, and sensor technologies that enhance the non-detrimental detection of food attributes. Among the key contributions are developing image-based analysis, hyper spectral imaging, and smart sensors for real-time quality monitoring. Conclusions The paper discusses different benefits and challenges identified in models by applying ML for better precision and efficiency while, on the other hand, having challenges like big data sets as well as model generalization generally. This means that ML will shortly transform and revolutionize fields concerned with the food quality assessment industry.

[12] addresses the critical issue of assessing the freshness of fruits and vegetables; this is as essential to quality and safety in food. The authors propose a novel classification approach utilizing deep learning techniques to improve the accuracy and efficiency of the detection of freshness. Mostly, the conventional methods use manual inspection or simple image processing that will eventually lead to subjective judgments and errors. The model developed shall thus apply enhanced image processing, deep neural network architecture with analysis of visual characteristics towards enhanced classification performance. Experiment results show the viability and possibility of the use of the developed model in real-time freshness assessment in agricultural and food industry processes. This study contributes to the advancement of food technological knowledge while emphasizing the need to integrate developing technologies for assurance of food quality and minimization of food waste in any portion of its supply chain.

[13] provides a comprehensive overview of all the machine learning techniques used in assessing fruit quality in the paper "Fruit Quality Evaluation using Machine Learning: A Review". The literature survey shows that there is a growing importance of automated systems in agriculture, especially in fruit quality assessment, with the general increase in demand for precision and speed. The authors discussed several ML algorithms such as SVM, ANN, and

KNN, for tasks of classification, grading, and defect detection in fruits, besides discussing numerous imaging technologies-integration with ML models, to improve the accuracy of quality evaluation. It is worth mentioning here that automation will also have the following advantages: there will be no human error in checking, consistency, and speed up of the process concerning checking the fruits. There are also associated challenges with model generalization, data acquisition and real-time deployment.

[14] provide a detailed review of fruit classification using deep learning-based models. The authors stated that proper fruit classification with high value is an important task in agriculture, retail, and food safety. Different deep learning models, which also include CNNs, are ready to go when it comes to image classification tasks where they sound well. The taken survey categorized the ways of fruit recognition, those showing development within model architectures, the invention of datasets, and also the role that data augmentation played in achieving better recognition accuracy. Besides, the authors appraise challenges inside the subject area: variability in the appearance of fruits, occlusion, and the need for strong models capable of acting in real conditions. More specifically, it places more emphasis on knowledge-sharing across domains and, above everything, proposes future research ideas, that is, hybrid models and perhaps fruit recognition systems integrated with IoT. Above all, this review will prove to be a great source of reference to those researchers and practitioners who have been tasked with the responsibility of making sense of the landscape presented by deep learning-powered fruit recognition technologies.

This paper [15] determines the fruit freshness in the food industry by image classification using transfer learning with CNN architecture. Residual networks performance is compared with the accuracy achieved while classifying freshness on six types of fruits, and residual networks have achieved accuracy above 99%. Against this backdrop, this paper used CNN models to test the fresh state of fruits within the food industry with reduction of background and image augmentation techniques.

This paper [16] with substantial details, AI techniques for defects, peculiarly related to the classification of fruit ripeness. Although applied machine learning and computer vision applied to CNNs were used, challenges in this respect included light variations, and one of the most urgent needs was large datasets in attaining progressive accuracy obtained during the study for detecting the level of ripeness. They further elaborate on transfer learning with an outline of future research, such as multispectral imaging integration. This paper would explain what the said innovation would do in terms of letting automation-with sound judgment within the field of agricultural handling-be able to produce proper fruit harvesting.

Here [17], different sensors used for catching data connected with freshness and spoilage is further enhanced through machine learning in terms of accuracy while analyzing. For sure, real-time monitoring and improvement of quality control are benefits, but challenges shall emerge through a better dataset and precision of the sensor. Application of these technologies shall be used to progress the assessment and evaluation of fish quality within the supply chain.

III. NATURAL PRODUCTS REVIEW

Quality review in natural products includes the consolidation of machines to classify them into classes, depending on the highlights they have. It, hence, includes forms such as picture preprocessing of the natural product, picture division, extraction of picture highlights, classification of natural regular items concurring to pictures, and machine-based identification. Figure 1 estimates the fruit quality.

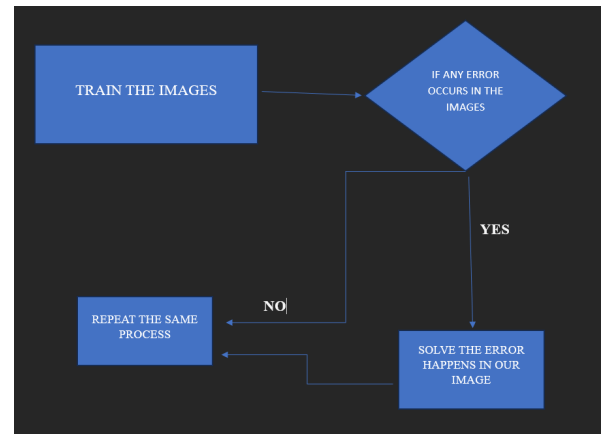


Fig 1 - Process of fruit quality estimation

We find out that the issue with this traditional machine comes to light on material failure of if it does not meet any given threshold, for instance. In other words, there is an issue with the component itself that makes it inherently problematic, such that defect detection inference there are very hard to realize whether visually or automatic inspection. Also, typically machine judgment is not able to say if the issue present in particular falls under or better may not fall major, minor, medium etc. For example, a small patch or scar will be normal in the black on the banana so this is not considered as defect but it may cause an orange to decay.

IV. PROPOSED METHODOLOGY

The Multi-Modal Convolutional Neural Network (MMCNN) algorithm combines both visual and spectral data for the classification of the quality and diseases of the algorithm. Initially It extracts spatial features from RGB images using Convolutional Neural Networks (CNN) and reduces spectral data dimensionality using Principal Component Analysis (PCA) to analyze the chemical properties like sugar, moisture content, etc. These features are fused into a single vector, capturing both visual and chemical properties. The model is then optimized using a Genetic Algorithm (GA) for hyper parameters, that further tune it and in turn enhance classifications accuracy, precision and reduce false predictions. The algorithm follows these steps to identify the quality, accuracy and precision etc.

A. Modern Technique

The calculation is Multi-Modal Convolutional Neural Arrange with Hereditary Optimization (MMCNN-GO). Here are the comes about displayed which is able tend to amalgamate a number of techniques towards superior classification of the apple illness. Subsequently, the demonstrate will take both RGB picture information and Near-Infrared (NIR) ghostly information to determine visual and chemical properties of the apple. So, when we consider highlights as spatial highlights from the imaging portion,

PCA is conducted on the ghastly information for highlight determination and dimensionality lessening. At that point these highlights are combined in a multi modal layers.

Step1: Data Collection and Preparation

Apples images (RBC) are collected to identify features like color, texture, and external defects.

$$\mathbf{I} \in \mathbf{R}^H \times \mathbf{W} \times \mathbf{C}$$

2. Collect NIR Spectral Data

$$\mathbf{S} \in \mathbf{R}^n$$

Step2: Feature Extraction from Image Using CNN

$$O(i, j) = \sum_{m=1}^k \sum_{n=1}^k \sum_{c=1}^c I(i+m, j+n, c) \cdot F(m, n, c)$$

Step3: Feature Extraction from Spectral Data Using PCA

$$\mathbf{S}_{\text{reduced}} = \mathbf{S} \times \mathbf{W}$$

Step4: Multi-Modal Feature Fusion

$$\mathbf{F} = [\mathbf{FCNN}; \mathbf{S}_{\text{reduced}}]$$

$$\mathbf{F} = \mathbf{F}_{\text{used feature vector.}}$$

Step5: Classification Using Fully Connected Layers and Soft max

1. Fully Connected Layer

$$\mathbf{Z} = \mathbf{W}_{fc} \cdot \mathbf{F} + \mathbf{b}_{fc}$$

2. Soft max Function

$$P(y = c | F) = \frac{\exp(Z_c)}{\sum_c \exp(Z_c)}$$

Step6: Optimization-Genetic Algorithm (GA)

$$f(\text{params}) = \text{Accuracy}(\hat{y}, y)$$

V. CHALLENGES

A. Optical and visual grading

Evaluation of the fruits is critical one and the system evaluates fruit by analyzing huge images of the same fruit for example apple, orange, banana etc. In order to have consistent accuracy. Evaluation is based on external characteristics, such as color and texture being principal factors but including also shape and size dimensions.

B. Methodology

In general, Newly Developed Advanced Technology Fruit Inspection Methods Recently, new methods have applied the developments in using cutting-edge technology that improve precision and accuracy. Using multispectral imaging It gives an integrated picture of fruit quality by capturing the various wavelengths that are used to evaluate the stages of decay in addition to maturity. What follows are some fruits that, with the application of segmentation

techniques like U-Net, need to be identified where certain health traits will have to be found. Feature engineering is one of the most important methods in incorporating relevant features. These are color, texture, and form that would enable evaluations of the general health of the fruit. Crowdsourcing approaches enable generalization, in which a large amount of labeled data is collected from numerous sources. Predictive modeling leverages environmental data among many others.

VI. RESULT AND DISCUSSION

Classification is a key feature when all quality assessment of food considered hence, it paved the way for hallucination in which human thinking is being lab induced to make judgments calls more correct and pulpier. They can perform the interpretation of images of fruits based on fundamental characteristics color, shape, size, texture and defects with the help of image processing tool. This is used while creating train set. Then, the knowledge base was tested for all training sets with Respective classification algorithms. Currently, there are few techniques to classify fruits. Those include KNN, ANN. Figure 2 shows the comparison analysis graph.

Two major visual indicators to identify an apple at a glance is its color and skin texture. As for apple skin color, it can range from green to a series of reds to yellow (all based on ripeness). This change in pigmentation is the main criteria from which degree of ripeness has been discerned and represent one off the initial properties that buyers as well as traders look for. Many researchers have studied in quality inspection of apples by image processing techniques. What I mean is, when it comes to color, apples we think of a red apple or a yellow apple, but as you see on the outside (not for example just fine) smoothness and texture says a lot about that object as edible, its quality levels and where its growth stage is. Attributes will distinguish that an apple is eatable or not. Figure 3 shows the sample apple dataset and figure 4 illustrates the apple with good quality.

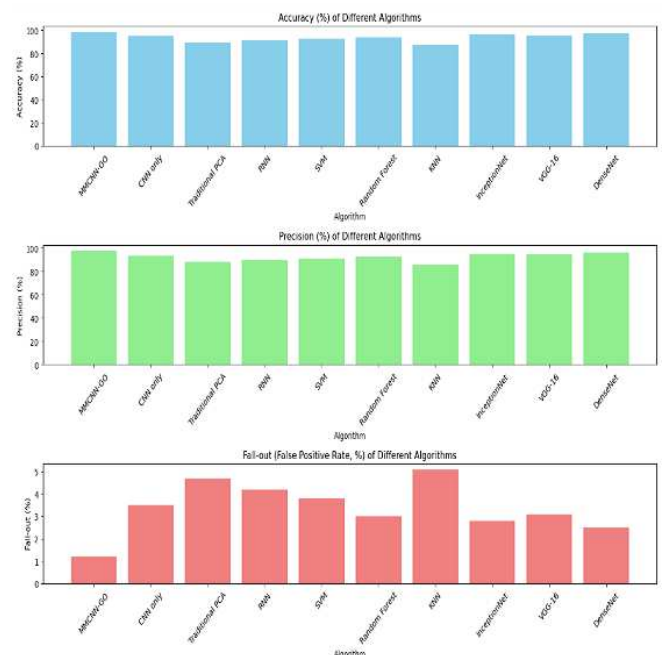


Fig 2 - Comparison analysis graph



Fig 3 Sample Apple dataset



Option:	1	2	3
			
Price:	\$4.00 / lb	\$1.50 / lb	\$1.00 / lb
Type:	Fuji	Gala	Gala
Freshness:	Average	Average	Poor

Option 1	Option 2	Option 3
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Fig 4 - Result with apple quality

VII. CONCLUSION

As such, it's imperative to sort the fruits and remove any bruises before selling them. This review study emphasizes the amalgamation of studies and categorization carried out by scholars. This endeavor aims to juxtapose several approaches that can be implemented within the system. These algorithms are used for fruit and other product classification. The KNN algorithm is crucial for this kind of classification since it determines the image's closest neighbor, or the image of the product that is needed to resemble the same product. Additionally, the product is sorted in order to retrieve the appropriate product. Shapes, sizes, colors, and textures are the fundamental

characteristics of fruits. The pre-processing, segmentation, and feature phases are fundamental to determining this.

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